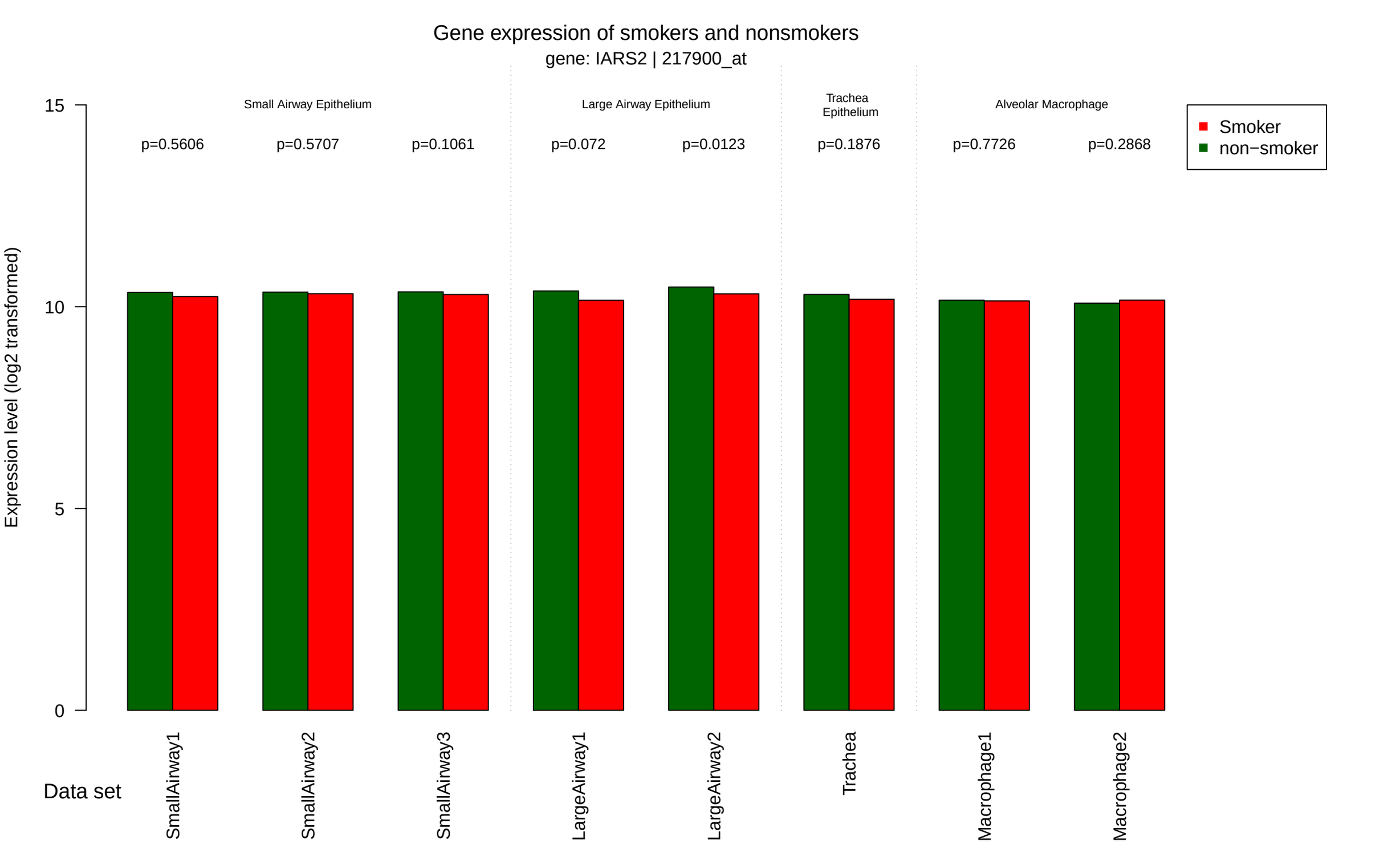
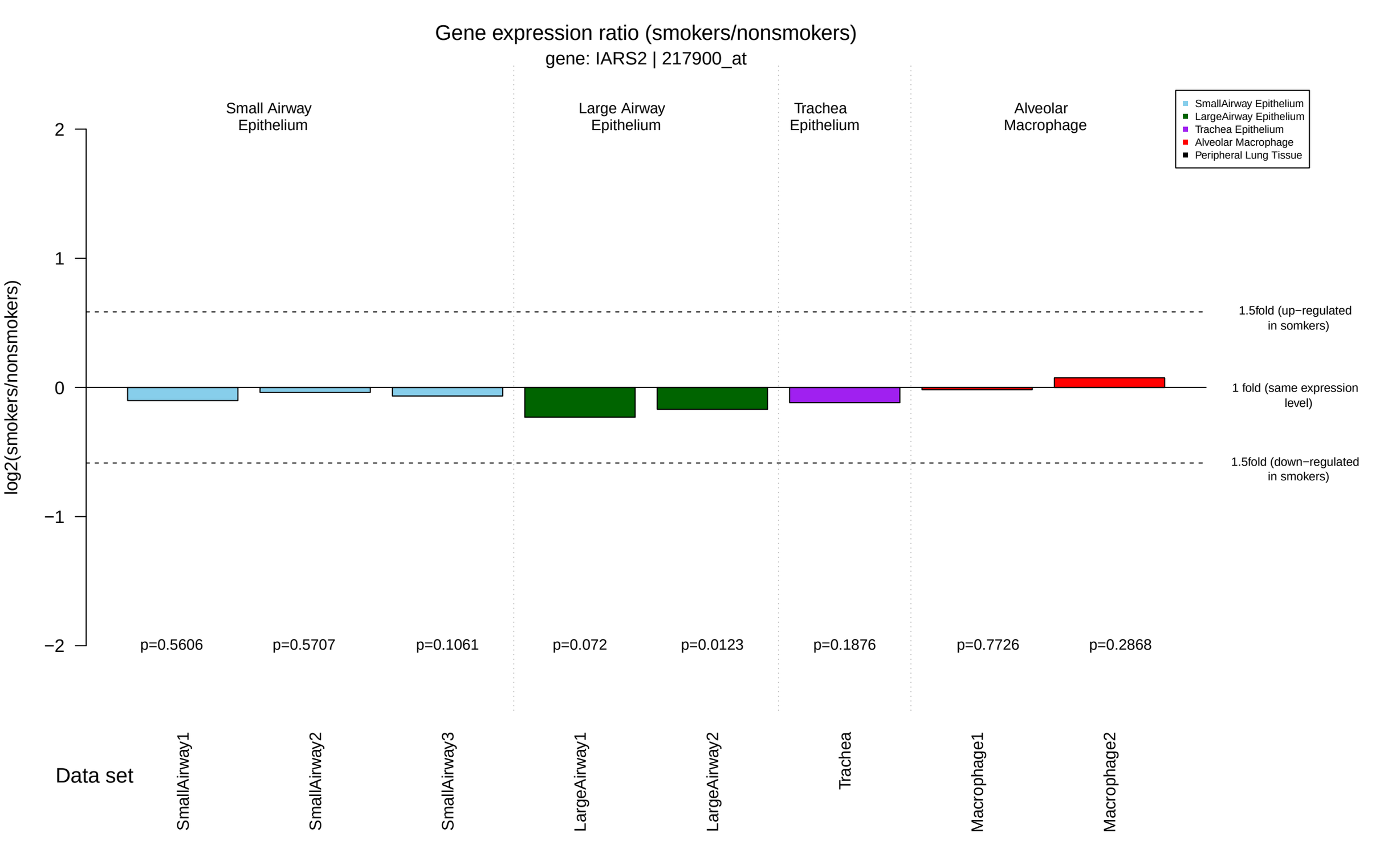
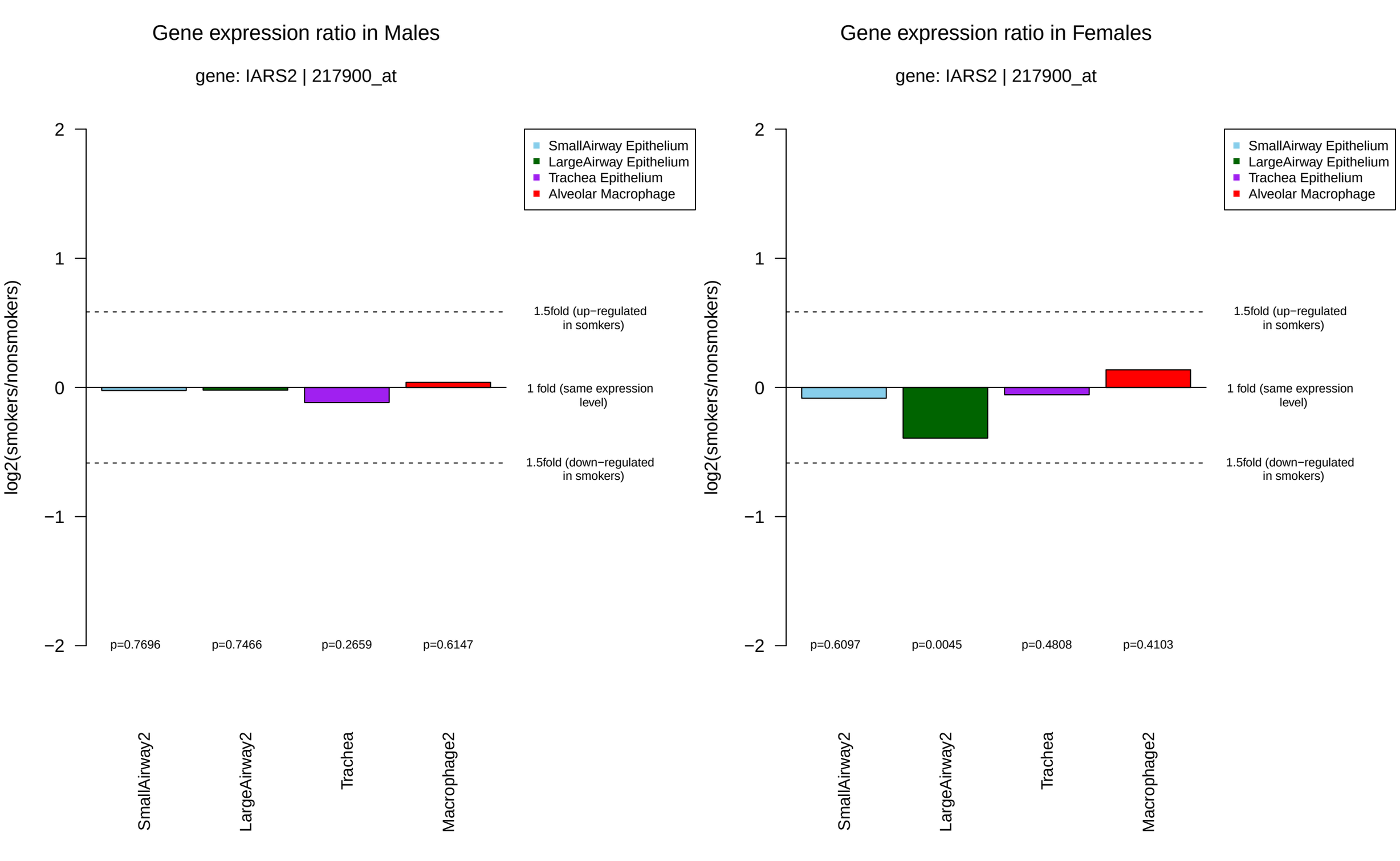
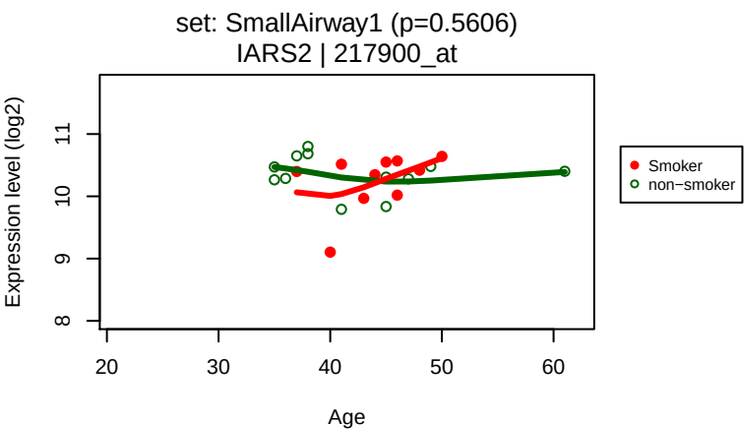
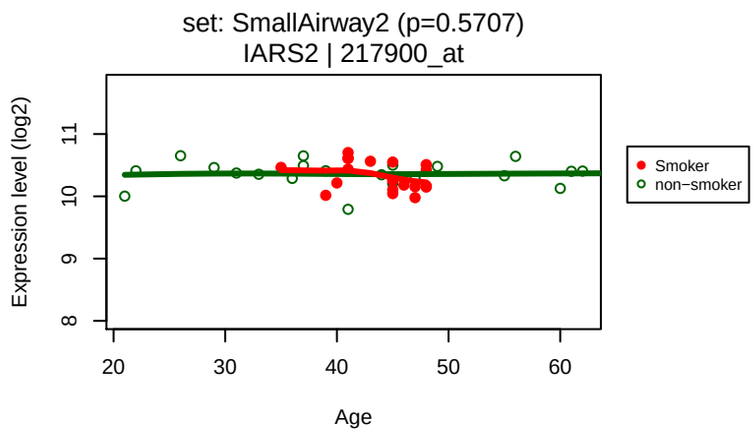
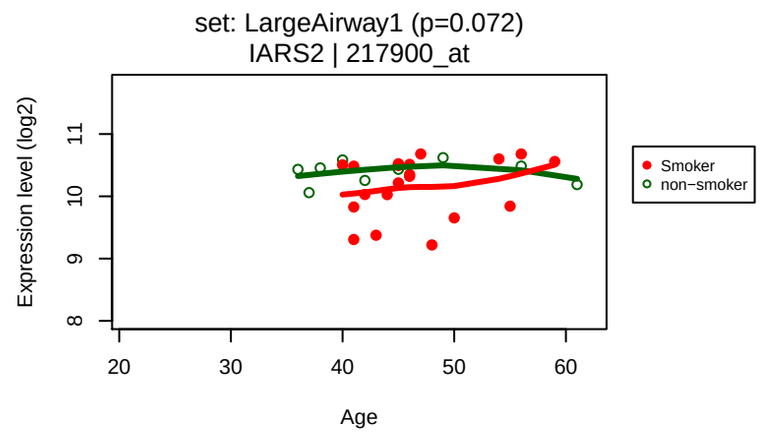

Gene expression of smokers and nonsmokers
gene: IARS2 | 217900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5606
p=0.5707
p=0.1061
p=0.072
p=0.0123
p=0.1876
p=0.7726
p=0.2868
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

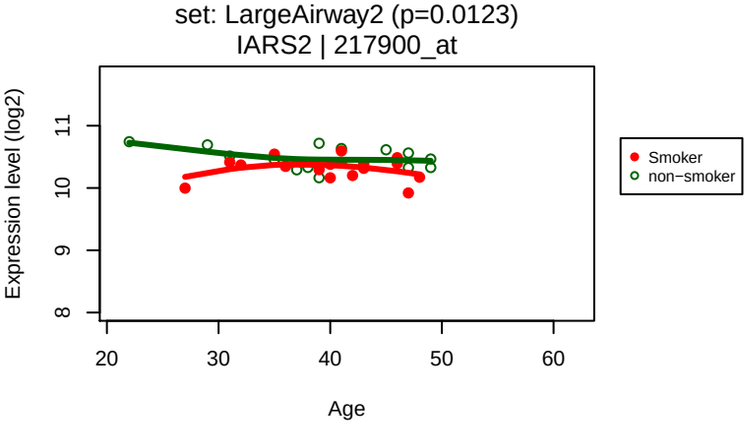

Gene expression ratio (smokers/nonsmokers)
gene: IARS2 | 217900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5606
p=0.5707
p=0.1061
p=0.072
p=0.0123
p=0.1876
p=0.7726
p=0.2868
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

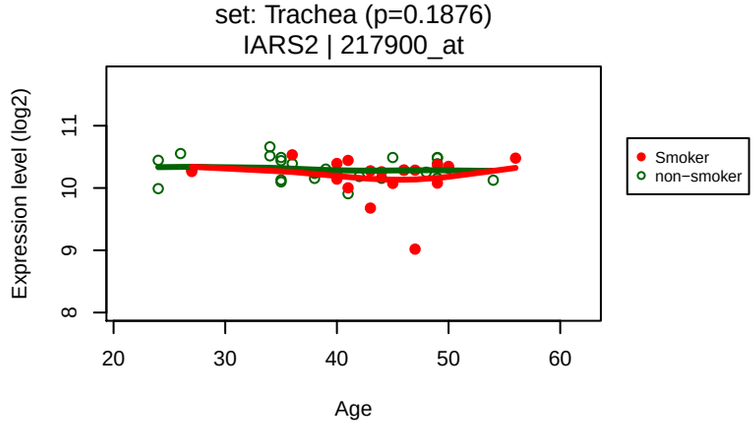

Gene expression ratio in Males
gene: IARS2 | 217900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7696
p=0.7466
p=0.2659
p=0.6147
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IARS2 | 217900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6097
p=0.0045
p=0.4808
p=0.4103
SmallAirway2
LargeAirway2
Trachea
Macrophage2

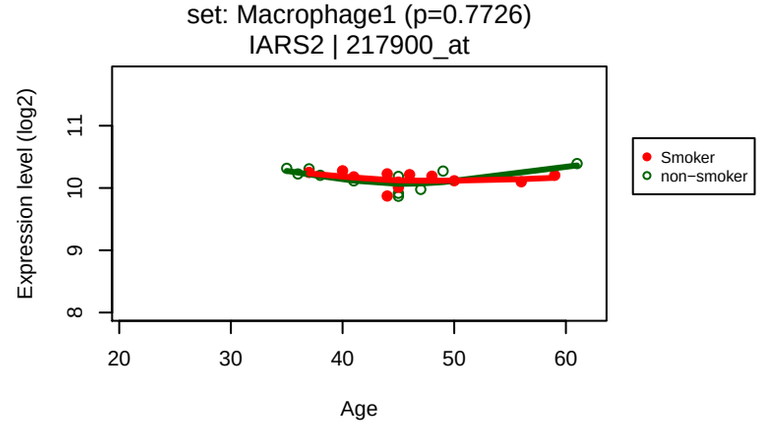

set: SmallAirway1 (p=0.5606)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker

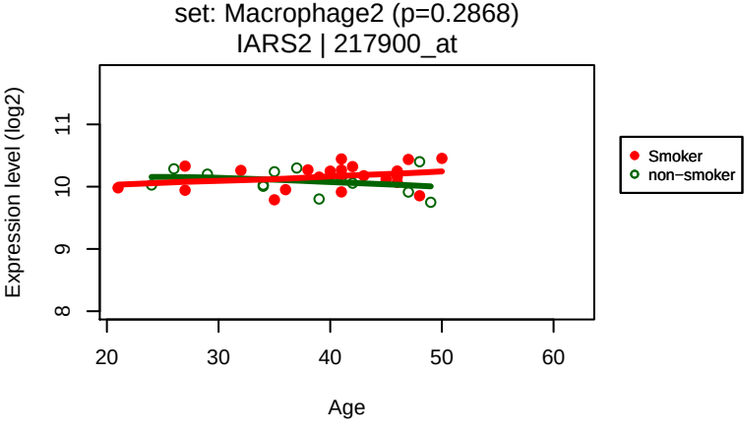

set: SmallAirway2 (p=0.5707)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker

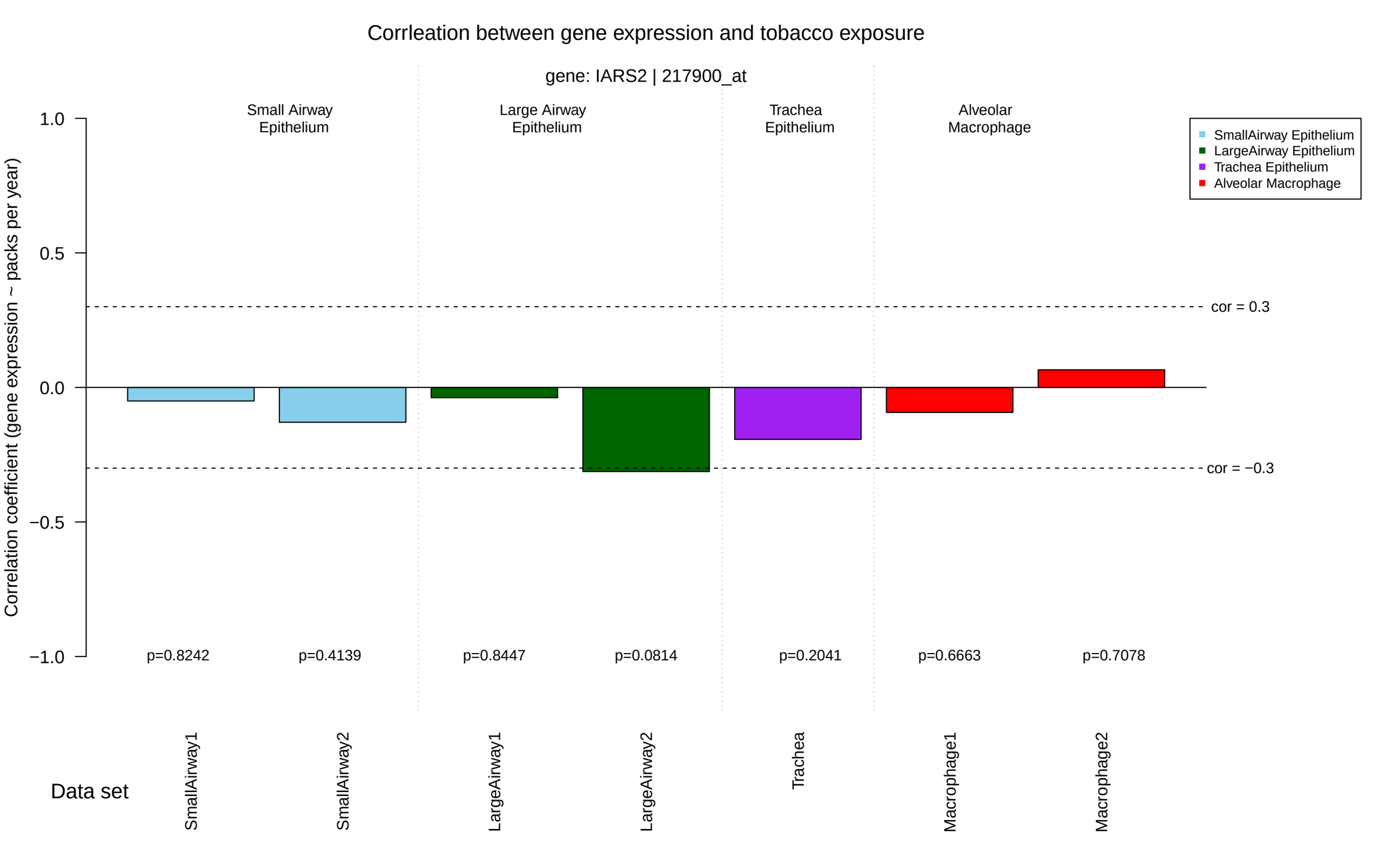

set: LargeAirway1 (p=0.072)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0123)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1876)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7726)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2868)
IARS2 | 217900_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IARS2 | 217900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8242
p=0.4139
p=0.8447
p=0.0814
p=0.2041
p=0.6663
p=0.7078
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2